

Adam MOUNIR

AI Research Student seeking a 5-6 months internship starting in April 2027
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EDUCATION

CentraleSupélec - Université Paris-Saclay

MSc. - Artificial Intelligence

Paris, FR
2026 - 2027

- **S1** : Optimization for ML - Decision Modelling - Machine Learning - Deep Learning
- **S2 Electives (MVA - ENS Paris-Saclay)** : Graph ML - Stochastic Optimization - Advanced Deep Learning - NLP - Reinforcement Learning

EFREI - Paris Panthéon-Assas

MEng. - Diplôme d'Ingénieur in Data & Artificial Intelligence

Paris, FR
2021 - 2026

- **Courses** : Convex Optimization - Stats/Probas - Graph Theory - Machine / Deep Learning - Reinforcement Learning - NLP

University of California, Irvine

Exchange Student, MEng. in Computer Science

California, USA
2023 - 2024

- **Courses** : Mathematics for Cryptography - Java Programming - Computer Architecture - Network Protocols

PUBLICATIONS

Neuron-Level Architecture Growth: A Controlled Evaluation for End-to-End EEG Decoding

IEEE ICASSP 2027 (under review) - A.Mounir, B.Aristimunha, S.Douka, S.Chevallier, A.Caillet

Sep 2026

Neuron-level growth for EEG decoders: networks widen during training instead of using a fixed width, reaching +2.9 pp over ShallowFBCSPNet at 0.57× parameters; the refusal rate of growth predicts when it helps.

PROFESSIONAL EXPERIENCE

Inria - TAU (Université Paris-Saclay)

AI Research - Intern

Paris, FR
Apr 2026 - Sep 2026

Researching deep-learning architectures for EEG decoding in [Braindecode](#) and Growing Neural Networks during training in [EEGrow](#)

- Design transformer, graph-neural-network, convolutional and Riemannian models for neural decoding using open EEG datasets
- Co-organize the NeurIPS 2026 [EEG decoding competition](#) (Meta, top academic labs) and building its benchmark infrastructure
- Porting in-training architecture growth ([gromo](#)) to EEG decoding: conv-junction width auto-sized during training rather than tuned

Société Générale

AI Engineer - Apprentice

Paris, FR
Oct 2025 - Apr 2026

Detected semantic discrepancies between unstructured document disclosures and structured multi-provider data

- Automated cross-source data alignment and causal influence tracing within hierarchical scoring models
- Engineered a Zero-shot NER pipeline using Claude 3.5 Sonnet (RAG) and Directed Graphs for topological impact analysis

Thales

AI Research - Intern

Paris, FR
Nov 2024 - Apr 2025

Researched global-scale Root Cause Analysis by distilling multi-step reasoning capabilities from LLMs into specialized architectures

- Designed a supervised alignment pipeline to map expert causal trajectories from logs into internalized reasoning schemas
- Distilled causal traces via GPT-4o to fine-tune a SFT pipeline using Llama3.1 (QLoRA) targeting latent reasoning convergence
- Achieved a 0.64 F1-score in RCA prediction and established a validation protocol for Out-of-Distribution scenarios

RESEARCH PROJECTS

Braindecode

Open-Source Contribution | [Github Link](#)

Maintaining and extending the reference PyTorch/skorch library for end-to-end neural time-series decoding

- Implementing transformer, graph-neural-network and Riemannian architectures for EEG/MEG signal decoding
- Building reproducible benchmarking pipelines over open neurophysiology datasets (MOABB integration)
- Contributing tested and documented modules to a community-driven library used across academic and clinical labs

Nullflow : Neuro-AI for Complementary Learning Systems

Personal Project | [Preprint Link](#)

Tackling catastrophic forgetting in Deep Neural Networks during non-stationary sequential data streams

- Designing bio-inspired architectures optimizing the stability-plasticity trade-off via CLS Theory & Null-Space Projection
- Built a Wake-Sleep/Flow Matching framework in PyTorch for NREM/REM latent generative replay (~2.1M parameters)
- Outperformed 8 CL baselines by +18.8pp via 6-step ODE latent replay

PERSONAL INTERESTS & SKILLS

Hobbies : Running, Swimming, Reading, Piano

Languages : French (native), English (C1), Spanish (B2)

Programming Skills : Python/PyTorch, C/Assembly 68000, Java/C#/Scala, MATLAB, HTML/CSS/JavaScript, Oracle, SQL